Sri Chaitanya IIT Academy.,India.

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A right Choice for the Real Aspirant

ICON Central Office - Madhapur - Hyderabad

Sec: Sr. XL-500, Sr.S60(N&S), Elite, IAS & LIIT **Paper -2(Adv-2025-P2-Model)** **Date: 25-03-2026**

Time: 02.30Pm to 05.30Pm **GTA-20** **Max. Marks: 180**

KEY SHEET

MATHEMATICS

1	**C**	2	**B**	3	**C**	4	**B**	5	**ABD**	6	**AD**
7	**BD**	8	**BCD**	9	**7**	10	**16**	11	**378**	12	**57**
13	**36**	14	**36**	15	**1**	16	**55**				

PHYSICS

17	**B**	18	**B**	19	**C**	20	**C**	21	**CD**	22	**CD**
23	**ABC**	24	**ABC**	25	**5**	26	**1**	27	**8**	28	**3**
29	**1**	30	**4**	31	**30**	32	**3**				

CHEMISTRY

33	**B**	34	**B**	35	**C**	36	**C**	37	**ABD**	38	**AC**
39	**ABC**	40	**D**	41	**12.68**	42	**0.5**	43	**12.43-12.46**	44	**0.97–0.99**
45	**4**	46	**21.9**	47	**11**	48	**70.8**				

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SOLUTIONS

MATHEMATICS

1. Let $\lim\limits_{n\to\infty}\prod\limits_{k=3}^{n}\left(1-\tan^4\frac{\pi}{2^k}\right)=L\Rightarrow L=\lim\limits_{n\to\infty}\cos\frac{\pi}{4}\prod\limits_{k=3}^{n}\left(\cos\left(\frac{\pi}{2^k}\right)\right)^{-3}$

$$\text{As } \frac{\sin x}{x}=\prod_{k=1}^{\infty}\cos\left(\frac{x}{2^k}\right)\Rightarrow\frac{\sin x}{x\cos\left(\frac{x}{2}\right)\cos\left(\frac{x}{4}\right)}=\prod_{k=3}^{\infty}\cos\left(\frac{x}{2^k}\right)$$

$$\Rightarrow\lim_{x\to\pi}\frac{\sin x}{x\cos\left(\frac{x}{2}\right)\cos\left(\frac{x}{4}\right)}=\lim_{x\to\pi}\prod_{k=3}^{\infty}\cos\left(\frac{x}{2^k}\right)\Rightarrow\prod_{k=3}^{\infty}\cos\left(\frac{x}{2^k}\right)=\frac{2\sqrt{2}}{\pi}\Rightarrow L=\frac{1}{\sqrt{2}}\left(\frac{\pi}{2\sqrt{2}}\right)^3=\frac{\pi^3}{32}$$

2. Put $x^5=t\quad I=\frac{1}{5}\int_0^1\frac{1+t^{2025}}{(1+t)^{2027}}dt$

$$=\frac{1}{5}\int_0^1\frac{1}{(1+t)^{2027}}dt+\frac{1}{5}\int_0^1\frac{t^{-2}}{\left(t^{-1}+1\right)^{2027}}dt$$

$$=\frac{1}{5}\times\frac{1}{2014}\quad\therefore\qquad p=5\times2026=2\times5\times1013$$

3. A) $x=t^2\qquad f(t)=t^3+\frac{1}{t^3}-4\left(t^2+\frac{1}{t^2}\right)$

$$f'(t)=3t^2-\frac{3}{t^4}-8t+\frac{8}{t^3}=\frac{3t^6-8t^5+8t-3}{t^4}\quad t=1,-1,\frac{3-\sqrt{5}}{2},\frac{3+\sqrt{5}}{2}$$

$$f_{minimum}=\left(t+\frac{1}{t}\right)^3-3\left(t+\frac{1}{t}\right)-4\left(t+\frac{1}{t}\right)^2+8=27-9-36+8\quad=-10\,for\,t+\frac{1}{t}=3$$

B) y = 2t $\qquad\int_0^4\frac{f'(y)e^{f(y)}dy}{2}=5\quad f(4)=\ln 11$

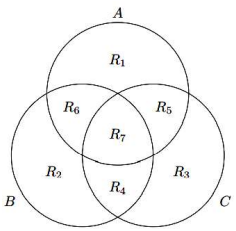
C) $x^n+\alpha x+\beta=(x-x_1)(x-x_1)(x-x_2)(x-x_3)----(x-x_{n-1})$

Differentiate with respect to x twice and substitute $x=x_1$

$$n(n-1)x_1^{n-2}=2(x_1-x_2)(x_1-x_3)----(x_1-x_{n-1})$$

D) $f(x)+f'(x)\le1\,e^xf(x)-e^0f(0)\le e^x-1$

4.



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Consider $N=|R_7|=|A\cap B\cap C|$

Case 1 : $N=2$

$|R_4|+N=|R_5|+N=|R_6|+N=2$

$|R_4|=|R_5|=|R_6|=0.$

$|R_1|+|R_5|+|R_6|+N=4$, so $|R_1|=2$. Similarly, $|R_2|=|R_3|=2.$

Number of ways $=\binom{8}{2}\binom{6}{2}\binom{4}{2}\binom{2}{2}=2520$ ways

Case 2 : $N=1$

$|R_4|+N=|R_5|+N=|R_6|+N=2$

$|R_4|=|R_5|=|R_6|=1.$

$|R_1|+|R_5|+|R_6|+N=4$, so $|R_1|=1$. Similarly, $|R_2|=|R_3|=1.$

Number of ways $=\binom{8}{1}\binom{7}{1}\binom{6}{1}.....\binom{2}{1}=40320$ ways

Case 3 : $N=0$

$|R_4|+N=|R_5|+N=|R_6|+N=2,$

$|R_4|=|R_5|=|R_6|=2.$

$|R_1|+|R_5|+|R_6|+N=4,$

$|R_1|=0$. Similarly, $|R_2|=|R_3|=0.$

Number of ways $=\binom{8}{2}\binom{6}{2}\binom{4}{2}=2520.$

Total number of ways = 2520 + 40320 + 2520 = 45360

5. $\sum_{r=0}^{n}\left(r^2-3r+1\right)C_r=2^{n-2}f(n) \quad \Rightarrow \sum_{r=0}^{n}\left[r(r-1)-2r+1\right]C_r=2^{n-2}f(n)$

$\Rightarrow \sum_{r=0}^{n}r(r-1)C_r-2\sum_{r=0}^{n}rC_r+\sum_{r=0}^{n}C_r=2^{n-2}f(n)$

$\Rightarrow n(n-1)2^{n-2}-2n\,2^{n-1}+2^n=2^{n-2}f(n)$

$\Rightarrow 2^{n-2}\left[n(n-1)-4n+4\right]=2^{n-2}f(n)$

$\Rightarrow f(n)=n^2-5n+4\Rightarrow f(x)=x^2-5x+4$

$\Rightarrow f(x)=(x-1)(x-4) \quad \Rightarrow f(x)=(x-1)(x-4)$

$\Rightarrow \alpha=4,\beta=1$ or $\alpha=1,\beta=4 \quad [\because f(x)=0]$

$\therefore 3\alpha+4\beta=12+4=16$ or $3+16=19$

Again $\alpha^2\beta+\beta^2\alpha=\alpha\beta(\alpha+\beta)=4(5)=20$

Also, $f(x)=x^2-5x+4 \quad \therefore f(x+1)=x^2-3x=x(x-3)$

$\therefore$ Roots of $f(x+1)=0$ and 3 and each root is 1 less than the roots of $f(x)=0.$

Now, $f(x-1)=0\Rightarrow(x-1)^2-5(x-1)+4=0$

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$\Rightarrow x^2-7x+10=0 \Rightarrow (x-2)(x-5)=0$

$\Rightarrow x=2,5=1+1,1+4=1+\alpha,1+\beta$

i.e., $\alpha+1,\beta+1$ are roots of $f(x-1)=0$

6. A number in decimal notation ends in a zero for each power of ten which divides it. Thus, we need to count both the number of 5s and the number of 2s dividing into our given expression. Since there are clearly more 2s than 5s, it is sufficient to count the number of 5s. One way to do this is as follows: 96 of the numbers 1!,2!,3!, 100! have a factor of 5. 91 have a factor of 10.86 have a factor of 15. And so on. This gives us an initial count of 96 + 91 + 86 +……..+1. Summing this arithmetic series of 20 terms, we get 970. However, we have neglected some powers of 5. every n! term for $n \geq 25$ has an additional power of 5 dividing it, for 76 extra; every n! for $n \geq 50$ has one more in addition to that, for a total of 51 extra; and similarly there are 26 extra from those larger than 75 and 1 extra from 100. Thus, our final total is 970+76+51+26+1=1124, and the answer is $\boxed{124}$.

7. $P=\frac{1}{2}\cos x\left[\sin(y+z)+\sin(y-z)\right]$

$\geq \frac{1}{2}\cos x.\sin(y+z)=\frac{1}{2}\cos^2 x$

But $x=\frac{\pi}{2}-(y+z)\leq \frac{\pi}{2}-2.\frac{\pi}{8}=\frac{\pi}{4}\Rightarrow P\geq \frac{1}{2}\left(\frac{1}{2}\right)=\frac{1}{4}$

Again, $P=\frac{1}{2}\cos z\left[\sin(x+y)-\sin(x-y)\right]$

$\Rightarrow P\leq \frac{1}{2}\cos^2 z=\frac{1+\cos 2z}{4}\Rightarrow P\leq \frac{1+\frac{1}{\sqrt{2}}}{4}=\frac{\sqrt{2}+1}{4\sqrt{2}}$

8. g is the inverse of p, hence p(a) = 0, p(b) =1

So f' is a continuous function which maps from $[0,1]\to[0,1]$

By intermediate value theorem we can say that there is a $k\in[0,1]$ such that $f'(k)=k$

$\int_0^1 f'(x)dx\leq p(b)-p(a)=1$

$f(1)-f(0)\leq 1$

By LMVT for p(x), $\frac{p(b)-p(a)}{b-1}=p'(c)$ for some $c\in(a,b)$

$\int_0^1 f'(x)dx\leq p(b)-p(a)\ \int_0^1 g'(x)dx=b-a\ \ \frac{\int_0^1 f'(x)dx}{\int_0^1 g'(x)dx}\leq p'(c)$

9. $A\cap(B\cup C)=$ There digit numbers which are of the form 9k + 2 or 9k + ℓ.

Sum of elements of $A\cap(B\cup C)$

$=\sum_{k=11}^{110}(9k+2)+\sum_{k=11}^{110}(9k+\ell)\ \ (if\ \ell>2)=(1091+\ell)100\Rightarrow \quad \ell=7$

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10. $A^2 = A.A = \begin{bmatrix} 2 & 3 \\ 0 & -1 \end{bmatrix}\begin{bmatrix} 2 & 3 \\ 0 & -1 \end{bmatrix} = \begin{bmatrix} 2^2 & 2^2-1 \\ 0 & 1 \end{bmatrix}$ $A^4 = \begin{bmatrix} 2^2 & 2^2-1 \\ 0 & 1 \end{bmatrix}\begin{bmatrix} 2^2 & 3 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 2^4 & 2^4-1 \\ 0 & 1 \end{bmatrix}$

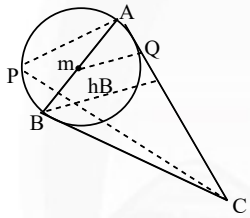
$A^{10} = \begin{bmatrix} 2^{10} & 2^{10}-1 \\ 0 & 1 \end{bmatrix}$ $2A = \begin{bmatrix} -4 & 6 \\ 0 & -2 \end{bmatrix}$ $\therefore adj(2A) = \begin{bmatrix} -2 & -6 \\ 0 & -4 \end{bmatrix} = -2\begin{bmatrix} 1 & 3 \\ 0 & 2 \end{bmatrix}$

$\therefore adj(2A)^{10} = 2^{10}\begin{bmatrix} 1 & 3 \\ 0 & 2 \end{bmatrix}^{10} = 2^{10}\begin{bmatrix} 1 & \left(2^{10}-1\right) \\ 0 & 2^{10} \end{bmatrix} = 2^{10}\begin{bmatrix} 1 & 1023 \\ 0 & 1024 \end{bmatrix}$

$A^{10} - (adj2A)^{10} = \begin{bmatrix} 0 & 2^n \times 1023 \\ 0 & 1-(1024)^2 \end{bmatrix}$ $\therefore \left|A^{10} - (adj2A)^{10}\right| = 0$

$\det\left(A^4\right) + \det\left(A^{10} - (adj2A)^{10}\right) = 16 + 0 = 16$

11. $AB = 10, BC = 17, CA = 21$



P lies on circle with $\overline{AB}$ as diameter to find greatest area of $(\Delta APC) \times 4$

P should be farthest from AC, on the circle .

Let M be the mid point of AB (in center of circle)

Q be foot from M on AC ,now $PQ = PM + MQ = \frac{1}{2}(AB) + \frac{1}{2}(h_B)$ (where h_B = length of altitude from B to AC)

But area of $\Delta ABC = \sqrt{24.14.7.3} = 84$

So $h_B = \frac{2(84)}{AC}$ $\left(\because \text{area} = \frac{1}{2}(AC)(h_B)\right)$ $= \frac{2\times 84}{21} = 8$

$\therefore PQ = \frac{1}{2}(10) + \frac{1}{2}(8) = 5 + 4 = 9$

$(GE)_{\text{greatest}} = 4\Delta ABC = 4\left(\frac{1}{2} \times 2 \times 9\right) = 2 \times 21 \times 9 = 378$

12. Consider $x = I + f, 0 \le f < 1$

$(I+f)^2 - af + b = 0 \Rightarrow I^2 + 2If + f^2 - af + b = 0$

$\Rightarrow f^2 + f(2I - a) + b + I^2 = 0$

$f_1 f_2 = b + I^2 \ge 1$ as $b \leftarrow \{1,2.....7\}$ $g(f) = f^2 + f(2I-a) + b + I^2$

$g(0) = b + I^2 >$

But $f < 1, f^2 + f(2I-a) + b + I^2$ should be less than zero at f = 1

$1 + 2I - a + I^2 + b < 0 \Rightarrow (I+1)^2 + b < a$ $(I+1)^2 < a - b \Rightarrow -6 < (I+1)^2 < 6$

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Possible value of I are $-3,-2,-1,0,1$ $\rightarrow I=-3\Rightarrow 4+b<a\Rightarrow b<a-4$

$a=7\Rightarrow b<3\{1,2\}$ $a=6\Rightarrow b<2\{1\}$ $a=5\Rightarrow b<1$

Number of solutions = I + 1 = 3

$I=-2\Rightarrow 1+b<a\Rightarrow b<a-1$

$a=7\Rightarrow b<6\{1,2,3,4,5\}$ $a=6\Rightarrow b<5\{1,2,3,4\}$

$a=5\Rightarrow b<4\{1,2,3\}$ $a=4\Rightarrow b<3\{1,2\}$

$a=3\Rightarrow b<2\{1\}$ $a=2\Rightarrow b<1$, Number of solutions = 5 + 4 + 3 + 2 + 1 = 15

$I=-1\Rightarrow 0+b<a\Rightarrow b<a\Rightarrow {}^7C_2=21$, $I=0\Rightarrow 1+b<a\Rightarrow$ same as $I=-2\Rightarrow 15$ solutions

$I=1\Rightarrow 4+b<a\Rightarrow$ same as $I=-3\Rightarrow 3$ solutions

Number of solutions = 3 + 15 + 21 + 15 + 3 = 57

13. Centroid $G\left(\frac{x_1+x_2}{3},\frac{y_1+y_2}{3}\right)$ $(\alpha,\beta)=\left(\frac{x_1+x_2}{3},\frac{y_1+y_2}{3}\right)$

$x_1+x_2=3\alpha, y_1+y_2=3\beta$ (x_1,y_1) & (x_2,y_2) are point 7

Ellipse $\Rightarrow \frac{x_1^2}{a^2}+\frac{y_1^2}{b^2}=1$ ____(1), $\frac{x_2^2}{a^2}+\frac{y_2^2}{b^2}=1$ ____(2)

$(1)-(2)\Rightarrow \frac{(x_1-x_2)(x_1+x_2)}{a^2}+\frac{(y_1-y_2)(y_1+y_2)}{b^2}=0$

$\Rightarrow \frac{(x_1-x_2)(3\alpha)}{a^2}+\frac{(y_1-y_2)(3\beta)}{b^2}=0 \Rightarrow$ slope of line $AB=\frac{y_2-y_1}{x_2-x_1}$

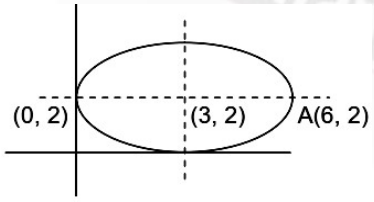
$m=-\frac{b^2\alpha}{a^2\beta}$ Consider line AB as $y=mx+c$ (x_1,y_1) and (x_2,y_2) lies on it

$\left.\begin{aligned}y_1&=mx_1+c\\ y_2&=mn_2+c\end{aligned}\right]\Rightarrow 3\beta=m(3\alpha)+2c \quad \Rightarrow c=\frac{(\beta-m\alpha)3}{2}$

Homogenizing ellipse with line $\frac{b^2x^2+a^2y^2}{a^2+b^2}-\left(\frac{y-mx}{c}\right)^2=0$

Right angle at origin $\Rightarrow \frac{1}{a^2}+\frac{m^2}{c^2}+\frac{1}{b^2}-\frac{1}{c^2}=0$

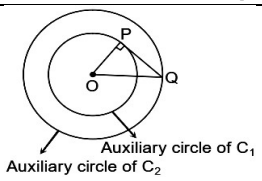
14. $C_1:\frac{(x-3)^2}{9}+\frac{(y-2)^2}{4}=1\therefore A_1=(6,2),A_2=(0,2)\therefore |PA_1-PA_2|=3\sqrt{2}$



Clearly locus of P is hyperbola for which $A_1A_2=2ae=6$ and $2a=3\sqrt{2}\Rightarrow e=\sqrt{2}$

$\therefore$ Locus of P is rectangular hyperbola. Equation of conic $C_2:(x-3)^2-(y-2)^2=\frac{9}{2}$

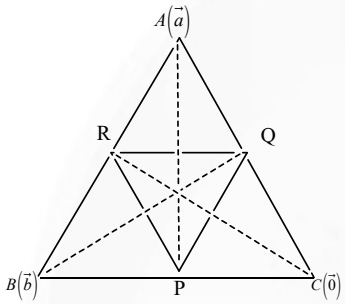
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Now, $D_1 = 2ae = 6$

$D_2 = b^2 = \frac{9}{2}$ $\quad D_3 : L = \sqrt{9-\frac{9}{2}} = \frac{3}{\sqrt{2}} \therefore \left(\frac{D_1 D_2}{D_3^2}\right)^2 = 36$

15. Let $C(\bar{0}), B(\bar{b}), \overrightarrow{OQ} = \frac{x\bar{a}}{x+z}, \overrightarrow{OG} = \frac{\bar{a}+\bar{b}}{3}$



P,Q,G collinear $\Rightarrow \exists \lambda, \mu, \delta$ such that

$\lambda\left(\frac{x\bar{a}}{x+z}\right) + \mu\left(\frac{y\bar{b}}{y+z}\right) + \delta\left(\frac{\bar{a}+\bar{b}}{3}\right) = 0$ _____(1)

$\lambda + \mu + \delta = 0$ _____(2)

$\bar{a} \| \bar{b} \Rightarrow \frac{\lambda x}{x+z} + \frac{\delta}{3} = 0$ _____(3)

$\frac{\mu y}{y+z} + \frac{\delta}{3} = 0$ _____(4)

(2), (3) + (4) $\Rightarrow \frac{(x+z)}{3x} - \frac{(y+z)}{3y} + 1 = 0 \Rightarrow \frac{x+z}{x} + \frac{y+z}{y} = 3$

$\Rightarrow \frac{z}{x} + \frac{z}{y} = 1 \Rightarrow \frac{1}{x} + \frac{1}{y} = \frac{1}{z} \Rightarrow \frac{2\Delta}{x} + \frac{2\Delta}{y} = \frac{2\Delta}{z} \Rightarrow h_a + h_b = h_c$

16. $\alpha x^{12} + bx^{11} + 1 = \left(x^2 - x - 1\right)\left(c_{11}x^{10} + c_{10}x^9 + \ldots\ldots + c_2 x + c_1\right)$

By comparison of LHS and RHS, we get

$c_1 = -1, c_2 = 1, c_3 = -2, c_4 = 3\ldots.$

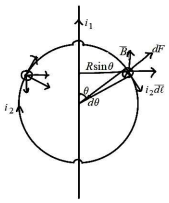
Which gives a pattern with alternate + and - signs and with each next terms magnitude is equal to the sum of magnitude of last two terms which is the property of Fibonacci series.

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PHYSICS

17. No. of SF will be 3.

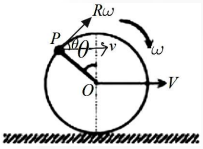


18. $B=\dfrac{\mu_0 i_1}{2\pi R\sin\theta}$ $\quad dF=i_2 d\ell B=\dfrac{\mu_0 i_1 i_2}{2\pi R\sin\theta}(Rd\theta)=\dfrac{\mu_0 i_1 i_2}{2\pi}\dfrac{d\theta}{\sin\theta}$

$$F=Fx=\int dFx=\int dF\sin\theta==\int_0^{2\pi}\frac{\mu_0 i_1 i_2}{2\pi}d\theta \qquad \therefore F=\mu_0 i_1 i_2$$

19. Negative of work done by friction is heat dissipated.

20. Volume of air bubble will not change.



21. $v+R\omega\cos\theta=0$ $\quad \cos\theta=-\dfrac{v}{R\omega}$ $\quad \Rightarrow\theta=\pi\pm\cos^{-1}\dfrac{v}{R\omega}$

22. $q=\varepsilon\sigma AT^4$ $\quad \lambda_m T=b$

23. Volume of conductor is equipotential.

24. $\bar{\tau}=\dfrac{d\bar{L}}{dt}=\bar{\omega}\times\bar{L}$

25. Apply maximum power transfer theorem by calculating thevenin resistance.

26. Net centripetal force must be equal to $\dfrac{mv^2}{r}$

27. In interval or 0 to 1/4 s, detector is active between 0 to 1/6 seconds.

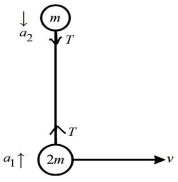
$\therefore$For 12 s, active time = $\dfrac{1/6}{1/4}\times 12=8s$

28. Apparent shift = $\Sigma t\left(1-\dfrac{1}{\mu}\right)$

29. $r=r_0 e^{-\lambda t}$ $\quad \dfrac{dr}{dt}=-\dfrac{r_0\lambda}{2}e^{-\lambda t/2}$ $\quad \therefore x_0=\dfrac{r_0\lambda}{2}=1$

30. $\alpha=\dfrac{MgL/2}{ML^2/3}=\dfrac{3g}{2L}$ and $\dfrac{Mg}{2}+Fy=\dfrac{M}{2}\times\dfrac{9g}{8}\Rightarrow Fy=\dfrac{Mg}{16}$

31. $a_2=2a_1$ $\quad a_1+a_2=\dfrac{v^2}{\ell}\Rightarrow a_1=\dfrac{v^2}{3\ell}$



$T=2ma_1=30N$

32. $\omega\cos\theta=v\sin\theta$ $\quad \omega=\dfrac{v}{l}\tan\theta=\dfrac{3v}{4\ell}$

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CHEMISTRY

33. Migration of phenyl ring occurs intramolecular way.

34. Fraction of body diagonal covered.

$$= \frac{2r + 2\times 0.414r + 4\times 0.225r}{\sqrt{3}\times\left(\frac{4r}{\sqrt{2}}\right)} = 0.76$$

$\therefore$ Fraction, not covered = 1 – 0.76 = 0.24

35. Statement (i) is incorrect because sulfur and other heavier chalcogens do not show +1 or −1 oxidation states, but the statement ignores the conditional nature of bonding environments.
Statement (ii) is correct: due to the inert pair effect, the +6 oxidation state of polonium is highly unstable and practically not observed.
Statement (iii) is correct: down the group, the +6 oxidation state becomes less stable while the +4 oxidation state becomes relatively more stable due to the inert pair effect.
Statement (iv) is incorrect because oxygen shows positive oxidation states in compounds such as OF_2.
Hence, only statements (ii) and (iii) are correct.

36. Threo configuartion

37. Side chains are oxideised to COOH.

38. For parallel first-order reactions: fraction unreacted = $x_Ae^{(-k_1t)} + x_Be^{(-k_2t)}$, where X_A and X_B are initial mole fractions.

39. Square species: $XeF_4(AX_4E_2), BrF_4^-(AX_4E_2), \left[Cu(NH_3)_4\right]^{2+}$ (square planar Cu^{2+}), $\left[PtC\ell_4\right]^{2-}$ (d^8 square planar). Hence, number of square species = 4.

Diamagnetic among square species: $XeF_4, BrF_4^-, \left[PtC\ell_4\right]^{2-}$. Thus, square and diamagnetic species = 3.

Tetrahedral and paramagnetic: $\left[FeC\ell_4\right]^{2-}$ (high-spin d^6) only.

$\left[Cu(NH_3)_4\right]^{2+}$ is square but paramagnetic $\left(d^9\right)$, so square and paramagnetic species = 1.

40. L = isopropylphenyl ketone

41. $n(A\ell) = \frac{0.270}{27} = 0.010\,mol$

$n\left(e^-\right)_{total} = 3\times 0.010 = 0.030\,mol$

$n\left(e^-\right)_{used} = 0.80\times 0.030 = 0.024\,mol$

$4e^- \rightarrow 8OH^- \Rightarrow 1e^- \rightarrow 2OH^-$

$n\left(OH^-\right) = 2\times 0.024 = 0.048\,mol$

$\left[OH^-\right] = \frac{0.048}{1.0} = 0.048\,M$

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$pOH = -\log(0.048) = -\log\left(48\times10^{-3}\right) = 3 - \log 48$

$\log 48 = 1.68 \Rightarrow pOH = 3 - 1.68 = 1.32$

$pH = 14 - pOH = 14 - 1.32 = 12.68$

42. Volume of platinum metal: $V = m/\rho = 1.00/20 = 0.05\,cm^3 = 5.0\times10^{-8}\,m^3$.

Total surface area of Pt particles: $A = 3V/r = 3\times\left(5.0\times10^{-8}\right)/\left(1.0\times10^{-6}\right) = 0.15\,m^2$.

Number of adsorbed hydrogen atoms: $N_H = A/a = 0.15/\left(5.0\times10^{-20}\right) = 3.0\times10^{18}$.

Moles of hydrogen molecules: $n(H_2) = \left(3.0\times10^{18}/2\right)/\left(6.0\times10^{23}\right) = 2.5\times10^{-6}\,mol$

Moles of electrons produced: $n\left(e^-\right) = 2\times2.5\times10^{-6} = 5.0\times10^{-6}\,mol$

Total charge: $Q = n\left(e^-\right)\times F = 5.0\times10^{-6}\times10^5 = 0.50\,C$

43. For this equilibrium at constant temperature: $a/(1-a)^2 \propto P$.

Hence, $P_2/P_1 = \left[a_2/(1-a_2)^2\right] \div \left[a_1/(1-a_1)^2\right]$

Given: $a_1 = 0.20, P_1 = 0.5\,atm$

$a_2 = 0.70$

Compute:

$a_1/(1-a_1)^2 = 0.20/(0.80)^2 = 0.3125$

$a_2/(1-a_2)^2 = 0.70/(0.30)^2 = 7.78$

Thus,

$P_2/0.5 = 7.78/0.3125 \approx 24.9$

$P_2 = 0.5\times24.9 \approx 12.45\,atm$

44. $RT = 0.08\times300 = 24\,L.atm.mol^{-1}$

$n_1 = \dfrac{PV_1}{RT} = \dfrac{1\times2.4}{24} = 0.10\,mol$

$n' = n_1 + 0.20 = 0.30\,mol$

$V' = \dfrac{n'RT}{P} = 0.30\times24 = 7.2L$

$W_1 = P(V' - V_1) = 1(7.2 - 2.4) = 4.8\,L.atm$

$W_2 = n'RTl\,n\left(\dfrac{V_2}{V'}\right) = 0.30(24)l\,n\,2 = 7.2(0.7) = 5.04\,L.atm$

$W = W_1 + W_2 = 4.8 + 5.04 = 9.84\,L.atm$

$W = 9.84\times0.1 = 0.984\,kJ$

45. Dry distillation of $Ca(CH_3COO)_2$ gives acetone (D).

Hence D is acetone. Since PCC oxidation of A also gives D, A must be 2-propanol.

A (2-propanol) $\rightarrow$ B (propene) [conc. H_2SO_4, Δ]

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B (propene) $\rightarrow CH_3-CHBr-CH_2Br\left[Br_2/CC\ell_4\right]$

$CH_3-CHBr-CH_2Br\rightarrow C\left(propyne, CH_3-C\equiv CH\right)\left[2\ eq\ NaNH_2\right]$

C (propyne) $\rightarrow$ D (acetone) $\left[dil.H_2SO_4, HgSO_4\right]$

Thus C is propyne $\left(C_3H_4\right)$.

$M\left(C\right)=M\left(C_3H_4\right)=3\times12+4\times1=40=Y\times10\Rightarrow Y=4$

46. Charcoal catalyzes the reaction: $SO_2+C\ell_2\rightarrow SO_2C\ell_2$.

Hence, A = SO_2 and B = $SO_2C\ell_2$.

Hydrolysis of sulfuryl chloride: $SO_2C\ell_2+2H_2O\rightarrow H_2SO_4+2HC\ell$.

For 0.10 mol B: H_2SO_4 = 0.10 mol $\rightarrow$ requires 0.20 mol $NaOH$

$HC\ell$ = 0.20 mol $\rightarrow$ requires 0.20 mol $NaOH$

Total $NaOH$ required = $0.40\,mol\rightarrow y=0.40\times40=16\,g$.

Reaction with phosphorus gives $C=PC\ell_3$.

$PC\ell_3+3CH_3COOH\rightarrow3CH_3COC\ell+H_3PO_3$

Thus, $D=CH_3COC\ell$ (molar mass = 78.5 $g\,mol^{-1}$).

7.85 g of D corresponds to 0.10 mol.

$CH_3COC\ell+2NH_3\rightarrow CH_3CONH_2+NH_4C\ell$

$E=CH_3CONH_2$ (molar mass = 59 $g\,mol^{-1}$).

$x=0.10\times59=5.9\,g$.

Therefore, x + y = 5.9 + 16 = 21.9.

47. A is the soluble argentothiosulfate complex: $A=\left[Ag\left(S_2O_3\right)_2\right]^{3-}$, hence x = 2.

On heating the intermediate silver thiosulfate, black precipitate C is Ag_2S. With $NaCN$/air it gives $D=\left[Ag\left(CN\right)_2\right]^-$ and $E=Na_2S$ (sulfide).

Sulfide ion gives the nitroprusside complex $\left[Fe\left(CN\right)_5NOS\right]^{4-}$, so $\alpha=4$ and $\beta=5$.

Zn displaces Ag from D, giving black $G=Ag\left(s\right)$. Ag dissolves in KCN (in air) to form the linear complex $\left[Ag\left(CN\right)_2\right]^-$, which has 3 linkage isomers: (C,C), (C,N), (N,N).

Hence y = 3.

Therefore, $\left(\alpha+\beta+x\right)$ / y = (4 + 5 + 2) / 3 = 11/3 = 3.67 (approx).

48. ΔH for $6Mn_2O_3\rightarrow4Mn_3O_4+O_2=+70.8kJ\left(\approx+71kJ\right)$.

Equivalently, for $4Mn_3O_4+O_2\rightarrow6Mn_2O_3, \Delta H=-70.8kJ$.

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